

Kartik Chandra

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EDUCATION

RV College of Engineering

B.E. in Computer Science and Engineering

- **CGPA:** 8.95 / 10.0
- **Coursework:** C, Python, Linear Algebra

Bengaluru, India

Sept 2024 – Aug 2028

EXPERIENCE

Technical Team Member – Astra Robotics

Robotics Projects and Research

- Working on **Bionic Arm** design using **McKibben Muscles and Pneumatic Actuation**.
- Developing **xArm – a 6DOF Robotic Arm** for **Mars Rover** manipulator control.
- Exploring **Neuromorphic control** techniques for intelligent actuator control, and implementation in Bionic Arm.

RVCE, Bengaluru

2024 – Present

Junior Executive – ACM Student Chapter

Tech Tank 2025 Hackathon Organizer

- Organized the flagship **Tech Tank 2025 Hackathon**, managing event operations, tech support and logistics.

RVCE, Bengaluru

2024 – Present

Member – Google Developer Groups (GDG)

Open Source & Dev, Competitive Programming Verticals

- Contributing to open source initiatives and internal developer tools. Focus on CP by creating weekly contests.

RVCE, Bengaluru

2025 – Present

PROJECTS

AI Research Agent

Python, FastAPI, Google Gemini 2.0 Flash, HTML/CSS/JS

- Built a **lightweight AI tool** to help create project ideas via fast summarization, insights, hypotheses, and citations.
- Designed a **6-second keyword-to-citation pipeline** using Gemini Flash and custom local processing.
- Developed both **backend and frontend** with an **offline-first**, minimal-latency architecture.

June 2025 – Present

Tennis Swing Classification and Analysis System

Python, FastAPI, Scikit-learn, Arduino, Web Dev

- Engineered a **real-time machine learning system** for sports analytics using a **Scikit-learn Stacking Classifier**.
- Deployed model on a **FastAPI server** with data from **MPU6050** IMU sensor and **MAX30102** HR Sensor.
- Achieved **85% accuracy**; built an interactive **web dashboard** using **HTML, CSS, JavaScript**.

Feb 2025 – Present

Predictive Maintenance System

Python, Streamlit, Hardware Interface

- Built a working prototype for **mechanical machine failure prediction** on conventional cutting, lathing and milling machines using sensor data, achieving **accuracy of 91%**.
- Implemented cloud integration and backend using **Python**.

Nov 2024 - Jan 2025

TECHNICAL SKILLS

- **Languages:** C, C++, Java, Python
- **Frameworks/Tools:** HTML, CSS, Streamlit, Scikit-Learn, Numpy, Pandas, Matplotlib
- **Hardware/Tech:** CAD - Solidworks 2022, Fusion 360; Microcontrollers - ESP8266, ESP32, Arduino UNO, R4-WiFi
- **Domains:** Machine Learning, Web Development, Competitive Programming, Embedded Systems, Robotics, System Design

ACHIEVEMENTS

Competitive Programming

2025 – Present

- **Codeforces rating:** 883; participated in global contests.
- **Leetcode rating :** 1387